

Vipul Yadav

Applied AI Engineer

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Profile

AI developer specializing in local LLMs, RAG pipelines, API integrations, and agentic workflows on Linux. Optimizes for both GPU and CPU inference. Bridges enterprise .NET/C# patterns with modern Python AI stacks, focusing on scalable, privacy-preserving ML.

Experience

Jan 2026 – **Applied AI Engineer**, ESUOR digital LLP, Mumbai

- May 2026
- Developing agentic AI workflows and multi-agent orchestration using **LangChain** and **LangGraph**.
 - Architecting robust **RAG pipelines** featuring structural document parsing via **PyMuPDF** and sub-second semantic retrieval using **ChromaDB**.
 - Implementing high-performance OCR solutions using **PaddleOCR**, optimized for **CPU-only inference** across Windows and Linux platforms.
 - Specializing in the deployment of local LLMs via **Ollama**, including custom resource-management fail-safes and model-switching logic for air-gapped environments.
 - Supporting multilingual text extraction for nearly all **Indian languages** with high precision.

Oct 2025 – **Data Scientist (Intern)**, ESUOR digital LLP, Mumbai

- Dec 2025
- Developed and tested data models for AI-driven applications using **Python** and machine learning libraries.
 - Gained hands-on experience in AI research and neural network foundations.

Jun 2025 – **Software Developer (Full-time & Intern)**, ESUOR digital LLP, Mumbai

- Sep 2025
- Built enterprise-level applications using C#, .NET Framework, and ASP.NET.
 - Managed database operations with PostgreSQL, MS SQL Server, Entity Framework and ADO.NET for scalable web engineering.
 - Applied software design principles and C++ for low-level performance optimization.

Featured Projects

- DocChat** Architected and deployed a high-performance, fully offline Document Intelligence platform enabling secure natural language querying of private PDF repositories.
- Engineered a custom **RAG pipeline** using PyMuPDF for structural parsing, SentenceTransformers for embeddings, and ChromaDB for semantic search.
 - Optimized local LLM orchestration via Ollama, implementing runtime model-switching and an automated **"resource-kill" fail-safe**.
 - Built a stateful conversation handler in **ASP.NET Core** using Session-based context management for multi-turn dialogue.
 - Impact:** Eliminated data privacy risks by migrating workflows to a 100% air-gapped environment with zero data leakage.

- NSE Historical Data Downloader** Developed a Python-based automation tool for bulk-downloading equity data from the National Stock Exchange of India for quantitative analysis.
- Bypassed advanced bot-detection layers through **reverse-engineering request flows** and implementing session rotation.
 - Engineered a robust scraping engine utilizing spoofed browser headers and randomized request delays to avoid rate-limit triggers.
 - Optimized data collection workflows for backtesting, enabling high-frequency retrieval for any NSE-listed equity.

- AI-Enhanced Student Management System** Built a full-stack student management platform using **C#, ASP.NET Core**, and **PostgreSQL** designed for high-efficiency administrative workflows.
 - Integrated **OpenAI GPT** and **DeepSeek APIs** to power an automated administrative chatbot for handling student queries, reminders, and enrollment assistance.
 - Developed a responsive frontend utilizing **Bootstrap** and **jQuery** for a seamless cross-device user experience.
 - **Impact:** Reduced average administrative query response times from hours to seconds through AI-driven automation.
- Multilingual OCR** Engineered a cross-platform text extraction system using **Python** and **PaddleOCR** optimized for CPU-only environments.
 - Optimized for diverse hardware architectures while generating structured JSON output for downstream processing.
 - Integrated support for nearly all major Indian languages with high-precision character recognition.

Technical Skills

- Languages Python (Advanced), C#, C++, SQL.
- AI/ML LangChain, LangGraph, Ollama, RAG, ChromaDB, SentenceTransformers, PyMuPDF, PaddleOCR.
- Web/Back-end FastAPI, ASP.NET Core, .NET Core 8 MVC, Next.js, Bootstrap.
- Databases PostgreSQL, MS SQL Server, Entity Framework, ADO.NET.
- Tools & Linux Linux Performance Tuning, Git.

Awards & Certifications

- 2025 **Winner, TechNova 2K25** – Intercollege Technical Quiz organized by Shree L. R. Tiwari Degree College.
- 2022 **Medhavi Chhatra Samman Purskar** – Issued by Government of Uttar Pradesh for state-level academic excellence.
- 2022 **RR Kabel Scholarship** – Selected as one of 500 students nationwide for academic merit.
- 2025 **Green Skills & AI Foundation** – Edunet Foundation certification.

Volunteering

- 2024–2025 NSS Media Co-Head @ Shree L.R. Degree College – Led media outreach and campaigns.
- 2025–Pres. Student Volunteer @ NSS UPES – Active involvement in environmental conservation projects.